ogy, GSM8K/SVAMP mathematics) remain isolated. We thus confirm that probes fail to generalize to new tasks in general but also that the situation is more nuanced. Indeed, this suggests that truthfulness representations adapt to domain-specific reasoning patterns rather than capturing one universal truth signal.

Underlying geometry. While previous work has documented the poor cross-task generalization of truthfulness probes, the geometric basis for this failure remains unexplored. We test whether generalization patterns reflect fundamental differences in how truthfulness is encoded across task domains, rather than mere statistical artifacts of limited training data. We examine cosine similarities between probe weight vectors (Figure 2). Most probe pairs exhibit near-orthogonal directions (cosine similarity < 0.5), with successful transfer occurring only between geometrically aligned probes. Indeed, Figure 3 shows that more similar probes also have better generalization performance in each others task (correlation coefficient of 0.59), showing that this is not just a mere geometrical problem but the cause for the performance drops; similar patterns with even lower cosine similarities are shown in the appendix for different models. These findings challenge the notion of a universal "geometry of truth." Instead, truthfulness directions emerge from task-specific representational structures, with different domains occupying orthogonal subspaces in the model's hidden state space. However, cosine similarity provides only a coarse measure of geometric relationships. To gain deeper insight into which specific features drive these differences, we next examine the support structure of sparse probes.

Sparse probes. To deepen our understanding of the features learned by probes, we consider sparse probes obtained through *sparse* logistic regression. Instead of Equation (1), we consider probes trained with L_1 regularization:

$$\min_{\substack{\theta \in \mathbb{R}^d, \\ b \in \mathbb{R}}} \frac{1}{N} \sum_{i=1}^{N} \log \left(1 + e^{-y_i(\theta^\top h_i + b)}\right) + \lambda_1 \|\theta\|_1 . \quad (2)$$

The ℓ_1-regularization coefficient λ_1 is tuned on a held-out validation set. As shown in Figure 4, this approach does not degrade performance. But it provides us with a visual way of comparing probes in Figure 5: The supports of probes between tasks are nearly disjoint. This can be made formal by computing support overlap, see Figure 6. We observe the same pattern as in Figure 2: some tasks have more support overlap than others and this coincides with the ones where probes generalize better.

Task-specific geometries. The previous experiments indicate that geometric information and generalization (failures) align with semantic task properties. This suggests that truthfulness detection mechanisms are not universal but

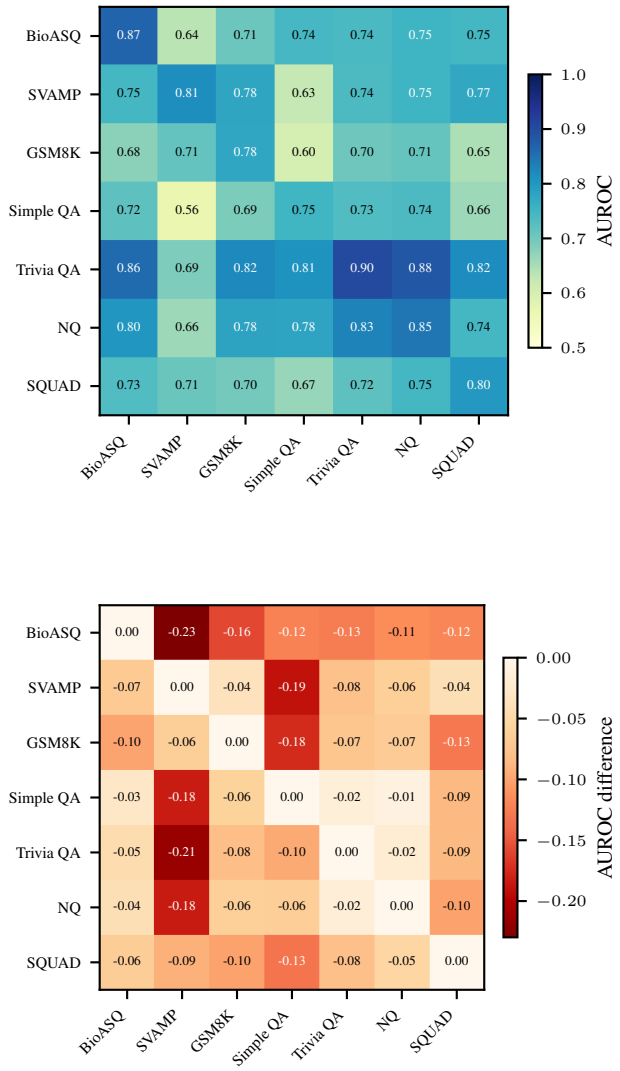

Figure 1. AUROC of probes trained on different tasks on the stop token of the output on last layer. Rows correspond to evaluation tasks while columns correspond to training tasks. The second plot represents the difference between the probe trained on this task and probes trained on the other datasets. Results are averaged over 5 runs.

rather emerge from task-specific representations within the model's hidden states. To visualize this phenomenon, we provide t-SNE plots (Van der Maaten & Hinton, 2008) of the hidden states of the model for each task Figure 7. What we see is that most of the tasks actually form distinct clusters. Similar tasks such as TriviaQA, NQ, SimpleQA are slightly mixed, but BioASQ or mathematical reasoning tasks are clearly separated from each other. The uncertainty signal, represented by the two right or wrong classes, is secondary compared to the distinction between tasks. This confirms

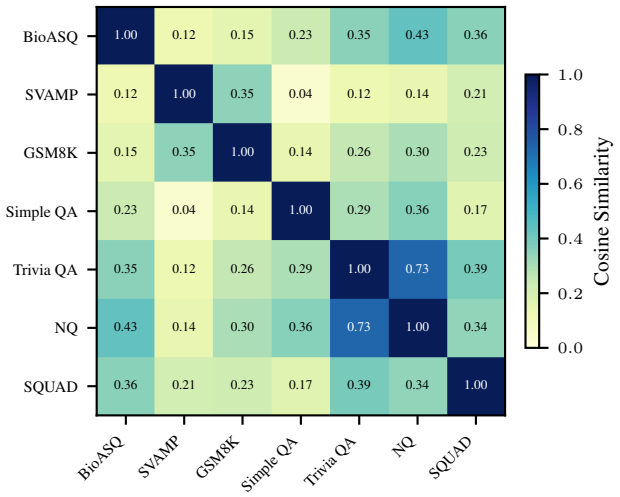

Figure 2. Cosine similarity between probes trained on different datasets using L2 regularization. Results are averaged over 5 runs. The cosine similarity between probe directions is consistently low (less than 0.5). Task pairs with large similarity Trivia QA - NQ (above 0.7) are the ones showing good generalization.

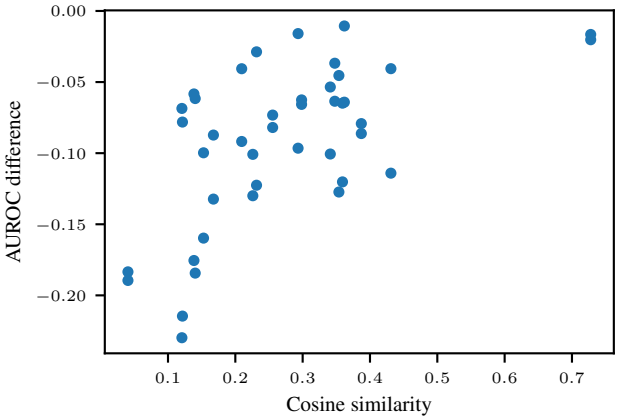

Figure 3. AUROC difference to probe trained on the right dataset as a function of cosine similarity between probes ($r = 0.59$, $R^2 = 0.35$, $p = 3.8 \times 10^{-5}$)

again our hypothesis that the truthfulness information is not universal and is very much task-dependent.

4. Generalization from mixture of tasks

4.1. Orthogonality in mixture of tasks

We now revisit our orthogonality hypothesis in a multi-task setting. We consider learning probes not on only one dataset but on a mixture of them, examining whether the

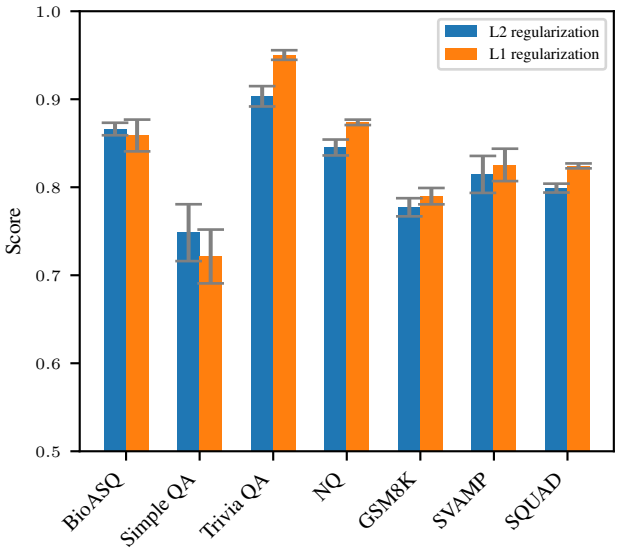

Figure 4. AUROC of linear probes with L1 or L2 regularisation. Results are averaged over 5 runs.

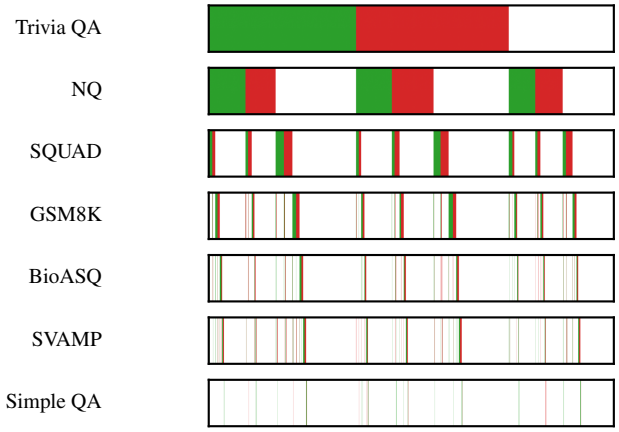

Figure 5. Signed support of sparse probes trained on different datasets at the stop token of the output, using L1 regularization at layer 28. Each row represents one probe trained on the corresponding dataset (y-axis labels). The x-axis shows the 3584 dimensions of the hidden state vector. Green indicates positive coefficients, red indicates negative coefficients, and white indicates zero coefficients (sparsity). Dimensions are sorted by sparsity level across all datasets, with the least sparse dimensions on the left and the most sparse on the right.

geometric relationships we observed persist in this more complex training scenario.

Figure 8 presents our main findings across several training strategies. The purple bars show the best performing probes: task-specific probes trained and tested on the same domain. The blue bars test our central question by showing performance when probes are trained on all tasks except the target

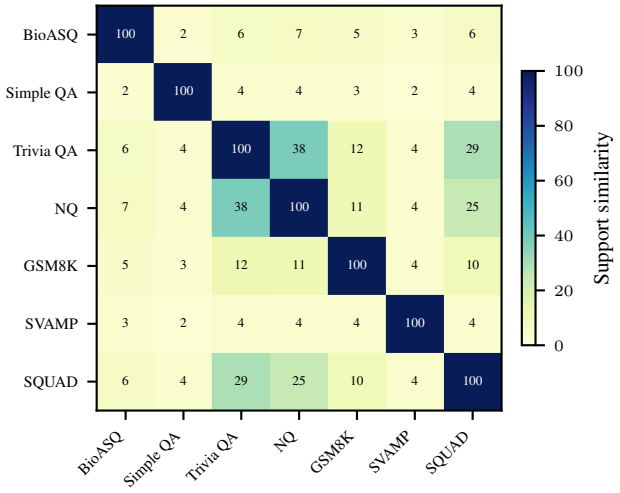

Figure 6. Support overlap between sparse probes trained on different datasets. Darker colors indicate higher overlap percentages. Task pairs with $> 30\%$ overlap (TriviaQA, NQ) correspond to successful cross-task generalization, while most pairs show $< 15\%$ overlap, explaining generalization failure.

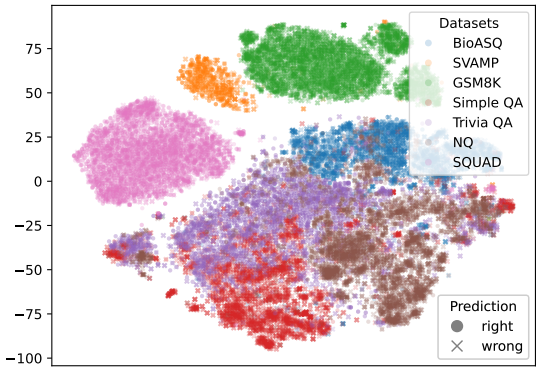

Figure 7. t-SNE plots of the hidden space at layer 28 at the stop token of the output. Different tasks form distinct clusters in representation space, with the correct/incorrect distinction being secondary to task boundaries.

task. Despite this diverse training mixture, we observe substantial performance drops across all domains, suggesting that multi-task training does not resolve the generalization problem.

To test whether this failure stems from our orthogonality hypothesis, we examine whether truthfulness directions from other tasks can be linearly combined to recover the direction from the target task. The orange bars show results from a constrained optimization where we restrict probe coefficients to lie within the subspace spanned by probes from other tasks. Formally, given probes $\theta_1, \ldots, \theta_6$ from non-target tasks, we solve:

$$\min_{\substack{\alpha \in \mathbb{R}^6 \\ b \in \mathbb{R}}} \frac{1}{N} \sum_{i=1}^{N} \log \left(1 + e^{-y_i(\theta_\alpha^\top h_i + b)}\right) + \frac{\lambda_2}{2} \|\alpha\|_2$$

$$\text{where } \theta_\alpha = \sum_{i=1}^{6} \alpha_i \theta_i \,.$$

The suboptimal performance of this constrained approach compared to task-specific probes confirms that target task directions lie outside the subspace generated by directions of the other tasks, providing direct evidence for our orthogonality claim.

The remaining comparisons in Figure 8 further support this interpretation. Training on all datasets simultaneously (red bars) yields performance roughly equivalent to simply summing individually trained probe parameters (green bars). This equivalence is striking: if probe directions overlapped significantly, naive parameter summation would cause destructive interference and degrade performance. Instead, the similar results confirm that probe directions are approximately orthogonal across tasks. However, both approaches still fall slightly short of task-specific training (purple bars), demonstrating that even when all tasks are included in training, the resulting probes cannot match the performance of domain-specific optimization.

4.2. Mixture of probes

Given the failure of linear probes to generalize across tasks, we test whether more complex probe architectures can improve cross-task transfer performance. We experiment with a "mixture of probes" approach inspired by Mixture-of-Experts architectures. Our approach uses a single gating layer with 16 expert probes, where each expert consists of a 2-layer feedforward network that takes the LLM's hidden states as input. The gating mechanism learns to route different inputs to different expert probes based on the hidden state representations.

For each target task, we train this mixture of probes on the six other tasks while using a mixture of these six tasks as a validation set. We perform grid search over hyperparameters including learning rate, weight decay, and auxiliary loss coefficients for the gating mechanism taking inspiration from Fedus et al. (2021).

Figure 9 presents results under two scenarios: an "oracle" setting where hyperparameters are selected using the test task performance, and a realistic setting where hyperparameters are chosen based on validation task performance.